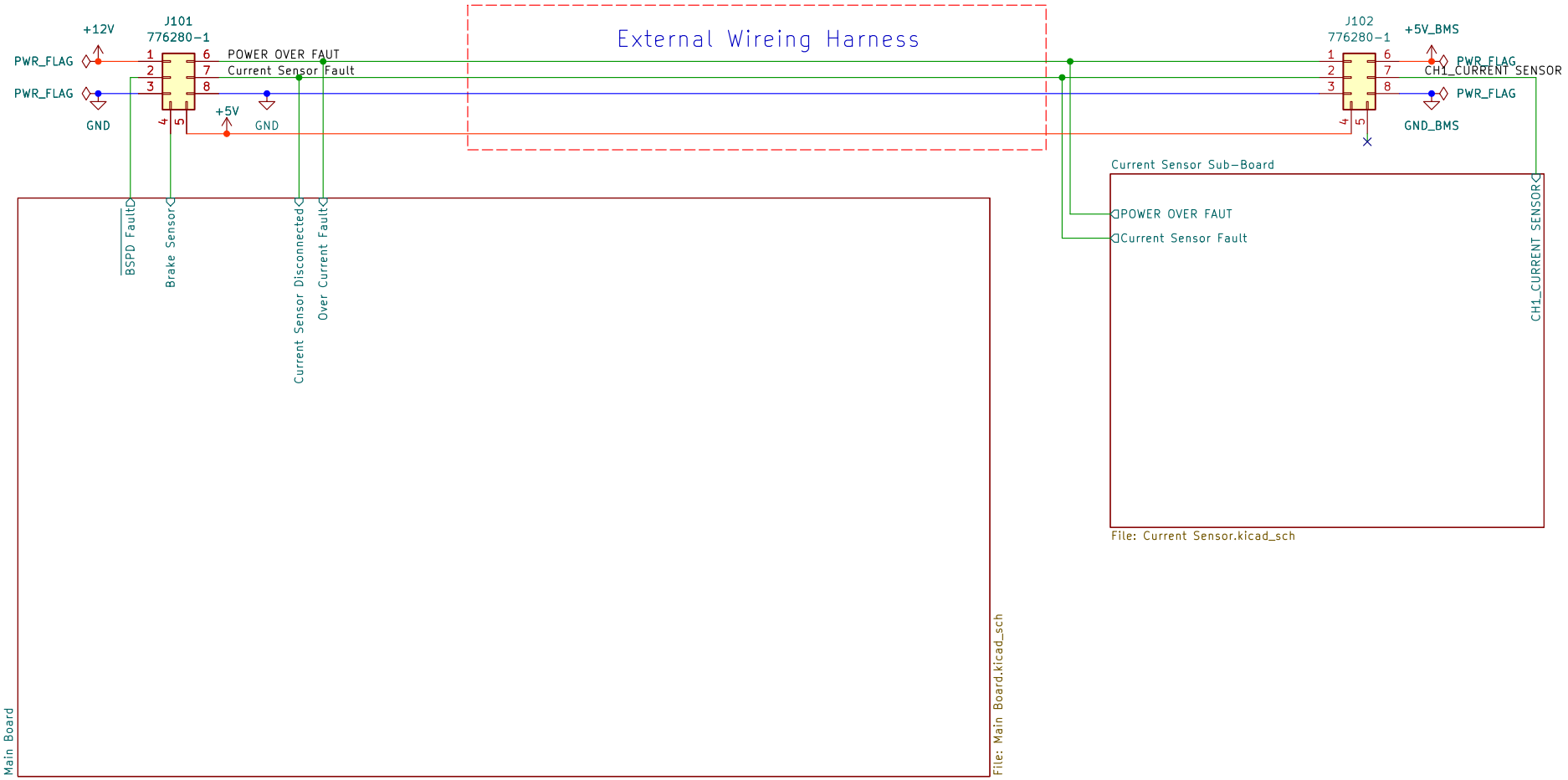


Current Sensor Test Wires -- Under Test, flow 1A over cable from Constant Current Power Source

J103
Conn_01x01_Socket
EV.5.2.5 Must Be rated for max Tractive System voltage

J104
Conn_01x01_Socket
Crimp connector to screw

15 Loops
Around sensor

If 15 loops
can not be made
adjust current
in wire

Batt. Nominal: 330V
Trip Power: 5kW
Trip Current: P = IV ⇒ I = P/V
I = 5000W/330V = 15.15A

Current per loop = I / n_{loops}
I_{test} = 15.15A/15 ≈ 1A

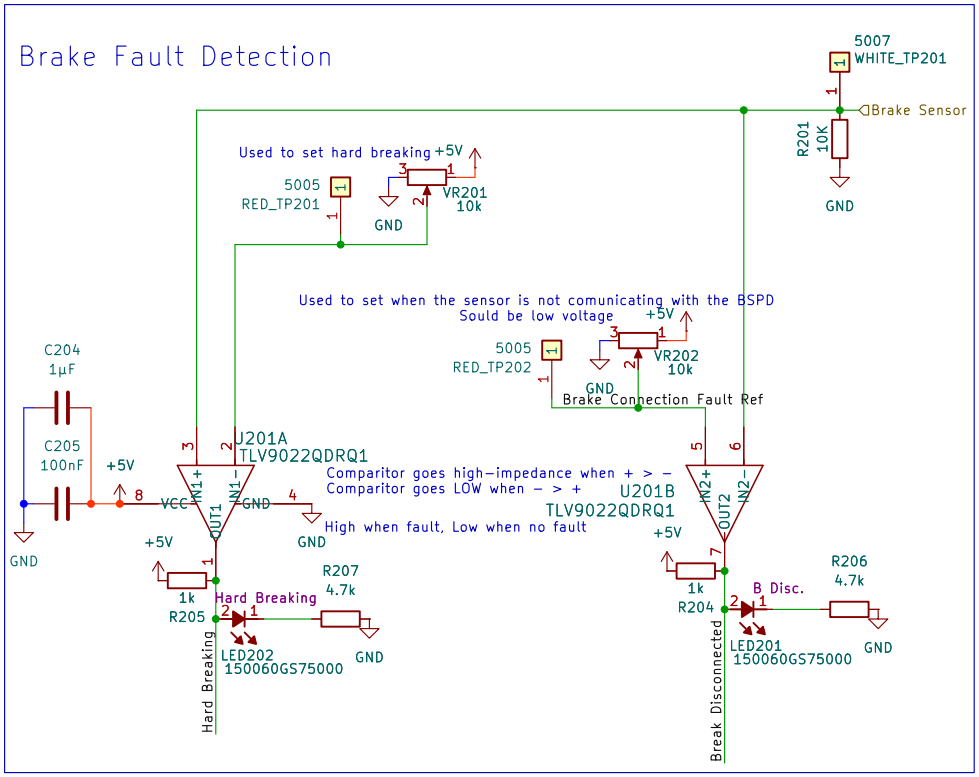
Dylan Shah
Carter Kortan

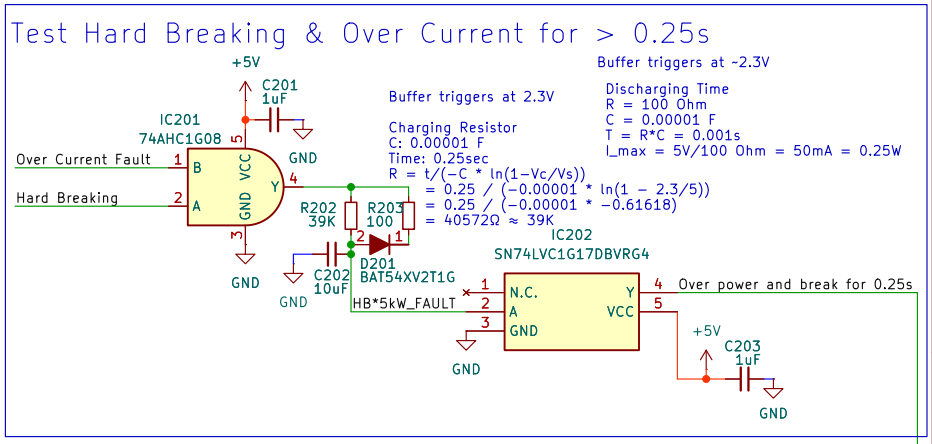
Sun Devil Motorsports

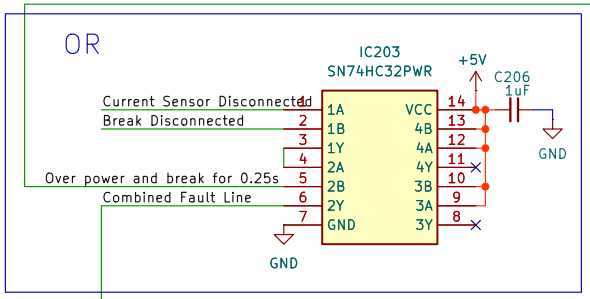
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File: sdm27ev-bspd-v2.kicad_sch

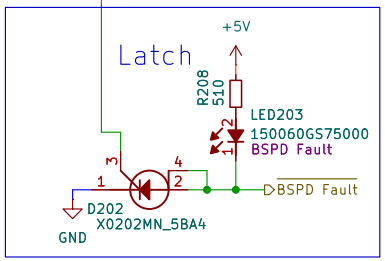
Title: BSPD

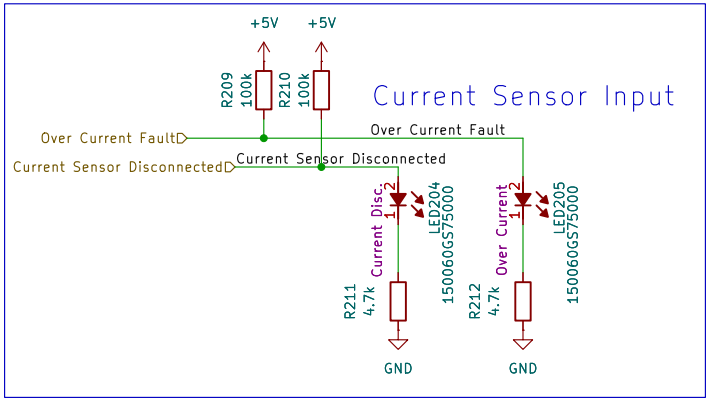
Size: USLetter	Date: 2026-10-24	**Rev: 2**
KiCad E.D.A. 10.0.7		Id: 1/3

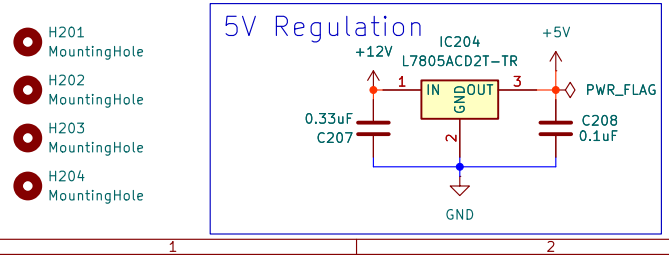












H201 MountingHole
H202 MountingHole
H203 MountingHole
H204 MountingHole

Dylan Shah
Carter Kortan

Sun Devil Motorsports

Sheet: /Main Board/
File: Main Board.kicad_sch

Title: BSPD Main Board

Size: USLetter	Date: 2026-10-24	**Rev: 2**
KiCad E.D.A. 10.0.7		Id: 2/3

This circutry is happning on the BMS side of the isolation between BMS and BSPD.
This is because the sensor has a max output of 1mA, and I could not find any way to isolate it without bringing BMS 5V here.
At that point, we might as well do the digital comparison beforehand, and then isolate the signal.

Used to set >5kW
Assuming this is a 330V battery, this should be ~3V

Used to set when the sensor is not comunicating with the BSPD
Sould be ~0.5V

+5V_BMS
5005
RED_TP301
VR301
10k
5007
WHITE_TP301
Over Current Ref
GND_BMS
CH1_CURRENT SENSOR
Under Current Ref
R301
10K
GND_BMS
+5V_BMS
5005
RED_TP302
VR302
10k
GND_BMS

Comparitor goes high-impedance when + > -
Comparitor goes LOW when - > +

C301
1µF
+5V_BMS
C302
100nF
GND_BMS
U301A
TLV9022QDRQ1
IN1+
IN1-
VCC
GND
OUT1
GND_BMS
U301B
TLV9022QDRQ1
IN2+
IN2-
OUT2

Over Current Fault
Current Sensor Fault
High Impedance when no fault
LOW when fault

+5V_BMS
R302
680
R303
680
+5V_BMS
U302A
LTV-827S
+5V
U302B
LTV-827S
+5V
R304
10K
R305
10K
GND
GND
R306
4.7k
R307
4.7k
P High Fault
P Low Fault
GND
LED301
150060GS75000
LED302
150060GS75000
GND
Brightness: 10mcd
ΔV = 5V - 2.7V = 2.3V
V/I = 2.3V / 0.5mA = 4600Ω
4.6k ≈ 4.7kΩ
LED Current Draw: 0.5mA
POWER OVER FAUT
Current Sensor Fault

Lines have 2.5mA when being driven high
Lines are HIGH on fault, and
LOW under normal opperation,
Put even larger pull up's on main board

BSPD SIDE

H301
MountingHole
H302
MountingHole
H303
MountingHole
H304
MountingHole

Dylan Shah
Carter Kortan

Sun Devil Motorsports

Sheet: /Current Sensor Sub-Board/
File: Current Sensor.kicad_sch

Title: BSPD – Current Sensor Sub Board

Size: USLetter	Date: 2026-10-24	**Rev: 2**
KiCad E.D.A. 10.0.7		Id: 3/3